Mathematics practice booklet: Grade 0 expectations

KG2 · Practice booklet

Name: _____

What this booklet is

This is general practice written for Mathematics at Grade 0 level, following the Common Core State Standards for Mathematics standards for that grade. It is not made from your child's MAP questions, and it does not replace the classroom. Use it as a guide to what a child is expected to know and do at this level.

How to use it

- Do one section at a time, about 10 minutes, 3 days a week. Stop while it is still enjoyable.
- Read each question all the way through, twice, before answering. Underline the key words and say what is being asked.
- Take your time. A careful, correct answer is worth more than a fast one.
- Use scrap paper and show the working. Then check the answer before moving on.
- Ask your child to explain how they got an answer, even when it is right.
- If a section feels too hard, do the section before it again, or ask the teacher. That is part of learning, not a problem.

Section 1: Patterns, equations and algebra thinking

Standards practised: K.OA.A.1, K.OA.A.2, K.OA.A.3, K.OA.A.4, K.OA.A.5

A worked example

Put 3 dates on a plate. Then put 2 more dates on the plate. How many dates are on the plate now?

1. Count the first dates: 1, 2, 3.
2. Add the 2 new dates and keep counting: 4, 5.
3. Putting 3 and 2 together makes 5.

5 dates

Questions

1. Maryam has 3 buttons. Her friend gives her 2 more buttons. How many buttons does Maryam have now?

   _____

2. There are 4 cups on the table. Salem puts 1 more cup on the table. How many cups are there?

   A. 3

   B. 5

   C. 6

3. Hamdan has 5 stones. He gives 2 stones to his brother. How many stones does Hamdan have left?

   _____

4. Noura has 2 red cups and 4 blue cups. How many cups does she have in all?

   A. 8

   B. 2

   C. 6

5. There are 7 dates in a bowl. Layla eats 3 of them. How many dates are left in the bowl?

   _____

6. Take 5 buttons. Show 2 ways to split them into two small groups. What are the groups?

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7. Khalid holds up 3 fingers. How many more fingers make 5?

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8. Omar has 6 stones. How many more stones does he need to make 10?

   A. 4

   B. 5

   C. 16

9. There are 10 cookies on a tray. Hind takes 4 cookies. How many cookies are left on the tray?

   A. 14

   B. 5

   C. 6

10. Look at the pattern: red button, blue button, red button, blue button, red button. Which button comes next?

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11. Look at the pattern: date, date, stone, date, date, stone, date, date. What comes next? How do you know?

Section 2: Shapes, measuring and space

Standards practised: K.G.A.1, K.G.A.2, K.G.A.3, K.G.B.4, K.MD.A.1, K.MD.A.2

A worked example

Which is longer: a new pencil or a small eraser? Put them side by side with the ends lined up at one end.

1. Line up the left ends of both things.
2. Look at the other ends. The one that sticks out more is longer.
3. Say the answer: the pencil reaches further, so it is longer.

The pencil is longer, because it reaches further

Questions

1. Maryam puts a round plate on the table. What shape is the plate?

   A. Square

   B. Circle

   C. Triangle

2. A cracker has 4 sides that are all the same length. What shape is it?

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3. Salem draws a triangle. How many sides does it have?

   A. 4

   B. 2

   C. 3

4. Noura has a rectangle tray. How many corners does the tray have?

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5. Hamdan kicks a ball. Is a ball a flat shape or a solid shape? What is its name?

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6. A tin of chickpeas is a solid shape. Which solid shape is it?
   A. Cylinder
   B. Cube
   C. Sphere

7. A ball is under the chair. Is the ball above the seat or below the seat?

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8. Maryam stands between Salem and Noura. Who is in the middle?

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9. Layla looks at a pencil and a date palm tree. Which is longer?
   A. A pencil
   B. A date palm tree
   C. A date

10. Hind lifts a watermelon and a date. Which is heavier?
    A. A watermelon
    B. A date
    C. They weigh the same

11. Omar's rope is longer than Layla's rope. Which rope is shorter?

Section 3: Data, graphs and chance

Standards practised: K.MD.B.3, K.CC.C.6, K.CC.C.7, K.CC.B.5

A worked example

On a tray there are 3 dates and 2 stones. Sort them into two groups. Which group has more?

1. Make one group of dates and one group of stones.
2. Count each group: dates 1, 2, 3 and stones 1, 2.
3. 3 is more than 2, so the dates are more.

The dates have more

Questions

1. Sort these into groups: 3 dates, 2 stones and 1 button. How many dates are there?

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2. There are 4 red cups and 3 blue cups. How many cups are there in all?

   A. 8

   B. 7

   C. 1

3. Salem has 5 stones and Hind has 3 stones. Who has more stones?

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4. Maryam has 4 buttons and Omar has 6 buttons. Who has fewer buttons?

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5. There are 4 dates and 4 cookies on a plate. Which sentence is true?

   A. There are more dates

   B. There are more cookies

   C. There is the same number of each

6. Sort the spoons and forks. There are 3 spoons and 4 forks. How many are there in all?

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7. There are 5 cups and 3 plates. How many more cups than plates are there?

   A. 2

   B. 3

   C. 8

8. There are 2 big stones and 5 small stones. Which group has fewer?

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9. There are 8 toys. 5 are cars and the rest are balls. How many balls are there?

   A. 13

   B. 3

   C. 4

10. 5 children want to sit down and there are 4 chairs. Is there a chair for every child? How many more chairs are needed?

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11. Sort 10 stones: 6 are smooth and 4 are rough. Which group has more, and how many more?

Section 4: Numbers and calculating

Standards practised: K.CC.A.1, K.CC.A.2, K.CC.A.3, K.CC.B.4, K.CC.B.5, K.CC.C.6, K.CC.C.7, K.NBT.A.1

A worked example

Count the dates in the bowl: there are 13. Show 13 as ten and some ones.

1. Make a group of 10 dates.
2. Count the dates left over: 1, 2, 3.
3. 13 is 10 and 3 ones.

13 is ten and 3 ones

Questions

1. Count out loud from 1 to 10. What number comes after 7?

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2. Maryam counts 12 dates one by one. When she says the last number, how many dates are there?

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3. Salem has 9 stones and finds 1 more. What is one more than 9?

   A. 10

   B. 8

   C. 11

4. Noura has 6 cups and puts 1 away. What is one less than 6?

   A. 7

   B. 4

   C. 5

5. Start at 8 and count on. Say the next three numbers.

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6. Count by tens: 10, 20, 30. What number comes next?

   A. 31

   B. 40

   C. 50

7. Which one is the number fifteen written with digits?

   A. 50

   B. 51

   C. 15

8. Salem has 8 stones and Noura has 6 stones. Which number is greater?

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9. Hamdan has 9 dates and Layla has 7 dates. Which number is smaller?

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10. 13 is ten and how many ones?

    A. 3

    B. 1

    C. 13

11. Ten buttons and 4 more buttons make what number? Which number comes just before 20?

Answer guide for parents

Section 1: Patterns, equations and algebra thinking

1. 5 (3 and 2 put together)
2. B. 5. 4 and 1 more is 5.
3. 3 (5 take away 2)
4. C. 6. 2 and 4 put together make 6.
5. 4
6. For example 1 and 4, or 2 and 3 (also 0 and 5).
7. 2
8. A. 4. 6 and 4 make 10.
9. C. 6. 10 take away 4 leaves 6.
10. A blue button
11. A stone (the pattern is date, date, stone, and it repeats).

Section 2: Shapes, measuring and space

1. B. Circle. A round plate is a circle.
2. A square
3. C. 3. A triangle has 3 sides.
4. 4
5. A solid shape, a sphere
6. A. Cylinder. A tin is a cylinder.
7. Below the seat
8. Maryam
9. B. A date palm tree. The date palm tree is much longer than a pencil.
10. A. A watermelon. The watermelon is heavier than a date.
11. Layla's rope

Section 3: Data, graphs and chance

1. 3
2. B. 7. 4 and 3 put together make 7.
3. Salem (5 is more than 3)
4. Maryam (4 is fewer than 6)
5. C. There is the same number of each. 4 and 4 are the same.
6. 7
7. A. 2. 5 is 2 more than 3.
8. The big stones
9. B. 3. 8 take away 5 leaves 3.
10. No, 1 more chair is needed.
11. The smooth stones, 2 more (6 is 2 more than 4).

Section 4: Numbers and calculating

1. 8
2. 12

3. A. 10. One more than 9 is 10.
4. C. 5. One less than 6 is 5.
5. 9, 10, 11
6. B. 40. Counting by tens, 30 and ten more is 40.
7. C. 15. Fifteen is a 1 for the ten and a 5 for the ones: 15.
8. 8
9. 7
10. A. 3. 13 is ten and 3 ones.
11. 14; the number before 20 is 19.